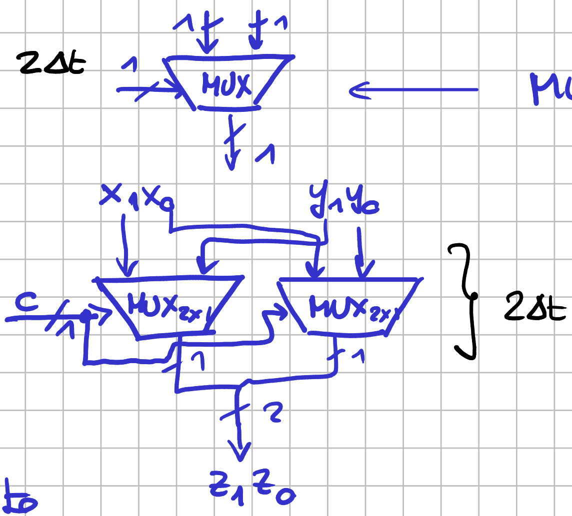
ESERCIZI

MUX



MUX 2x1

↑ n° ingressi ↑ n° di bit

MUX 2x2

↓

fa passare sull'uscita l'ingresso da 2 bit scelto dall'ingresso di controllo

1) TAB VERITÀ

c x_1x_0 y_1y_0	z_1z_0
0 x_1x_0 - -	x_1 x_0
1 - - y_1y_0	y_1 y_0

c	x_1	x_0	y_1	y_0	z_1	z_0
0	0	0	-	-	0	0
0	0	1	-	-	0	1
0	1	0	-	-	1	0
0	1	1	-	-	1	1
1	-	-	0	0	0	0
1	-	-	0	1	0	1
1	-	-	1	0	1	0
1	-	-	1	1	1	1

$$\bar{c}x_1(\cancel{\bar{x}_0 + x_0})$$

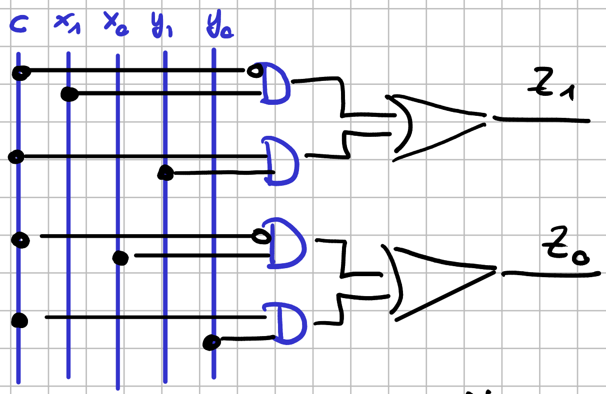
$$z_1 = \bar{c}x_1\bar{x}_0 + \bar{c}x_1x_0 + c y_1\bar{y}_0 + c y_1 y_0 =$$

$$c y_1(\cancel{\bar{y}_0 + y_0})$$

$$z_1 = \bar{c}x_1 + c y_1$$

$$z_2 = \bar{c}\bar{x}_1x_0 + \bar{c}x_1x_0 + c y_0$$

$$z_0 = \bar{c}x_0 + c y_0$$



Δt + Δt

c	x_1	x_0	y_1	y_0	z_1	z_0
0	0	0	–	–	0	0
0	0	1	–	–	0	1
0	1	0	–	–	1	0
0	1	1	–	–	1	1
1	–	–	0	0	0	0
1	–	–	0	1	0	1
1	–	–	1	0	1	0
1	–	–	1	1	1	1

z_1

C=0

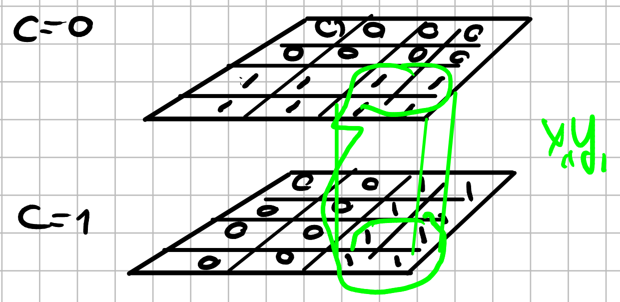
x_1x_0 \ y_1y_0	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

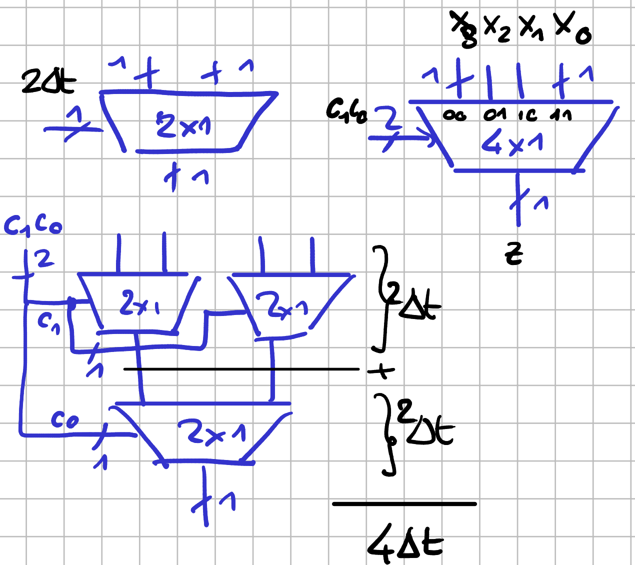
$\bar{c}\,x_1$

C=1

x_1x_0 \ y_1y_0	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	0	1	1

$c\,y_1$





c_1	c_0	x_3	x_2	x_1	x_0	z
0	0	1	–	–	–	1
0	0	0	–	–	–	0
0	1	–	1	–	–	1
1	0	–	–	1	–	1
1	1	–	–	–	1	1

← 0 ⇒ no termine AND

6 ingressi

termini AND da 3 ingressi

$$z = \bar{c}_1\bar{c}_0 x_3 + \bar{c}_1 c_0 x_2 + c_1\bar{c}_0 x_1 + c_1 c_0 x_0$$

quanti livelli di logica? (con porte da 2 ingressi)

$2 = \lceil \log_2 4 \rceil$ ← OR di 4 ing ciascuno

$+$

$2 = \lceil \log_2 3 \rceil$ ← AND di 3 ing

$4\Delta t$

MUX 2x1

c	x	y	z
0	1	–	1
1	–	1	1

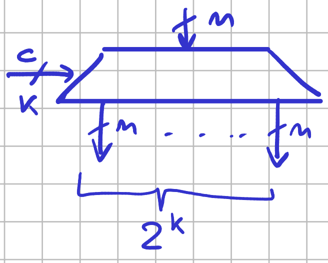
$$z = \bar{c}\,x + c\,y$$

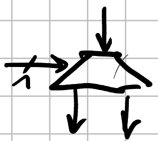
OR da 2 ing

ciascuno AND da 2 ing

$2\Delta t$ con

DEMUX





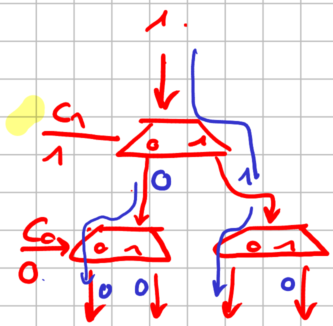
c	x	z_1	z_0
0	1	1	0
1	1	0	1

$z_1 = \bar{c}x$

$z_2 = cx$

1 liv logico

$1\Delta t$



$2\Delta t$

VS

DEMUX 1bit su 4 uscite

c_1	c_0	x	z_3	z_2	z_1	z_0
0	0	1	1	0	0	0
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	1

$z_3 = \bar{c}_1\bar{c}_0 x$

$z_2 = \bar{c}_1 c_0 x$

$z_1 = c_1\bar{c}_0 x$

$z_0 = c_1 c_0 x$

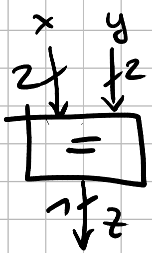
porte da 2 ingressi

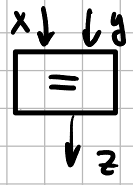
$\bar{c}_1\bar{c}_0 x$

2 livelli di logica (AND)

$2\Delta t$

CONFRONTATORE





$x = y \Rightarrow z = 1$

else $z = 0$

$x_1 x_0 y_1 y_0$	z
00 00	1
01 01	1
10 10	1
11 11	1

x y	z
00	1
01	0
10	0
11	1

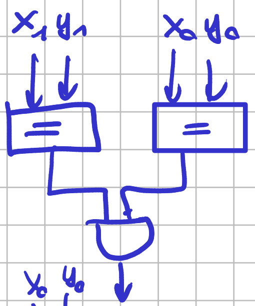
$z = \bar{x}\bar{y} + xy$

2 livelli di logica $(2\Delta t)$

$$z = \overline{\bar{x}_1 \bar{x}_0 \bar{y}_1 \bar{y}_0 + \bar{x}_1 x_0 \bar{y}_1 y_0 +}$$
$$x_1 \bar{x}_0 y_1 \bar{y}_0 + x_1 x_0 y_1 y_0$$

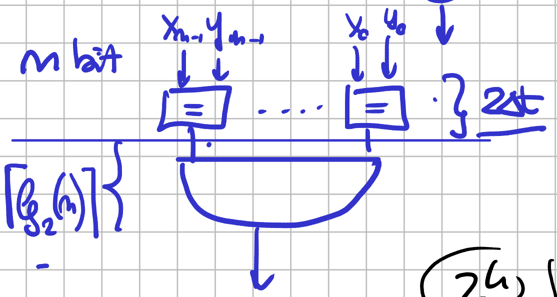
4 livelli di logica

2bit



liv di logica ? x n bit

n bit



TV

n+n	z
	2^n "1"

n = 4

$x_3 x_2 x_1 x_0$	$y_3 y_2 y_1 y_0$	z
0000	0000	1
0001		1
0010		1
0011		1

2n

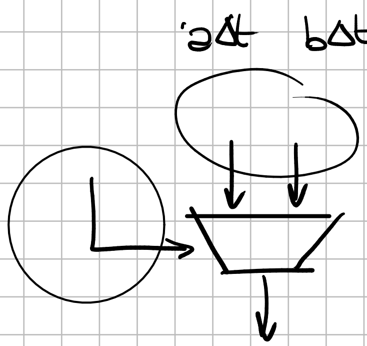
2^4 16

2^n "1"

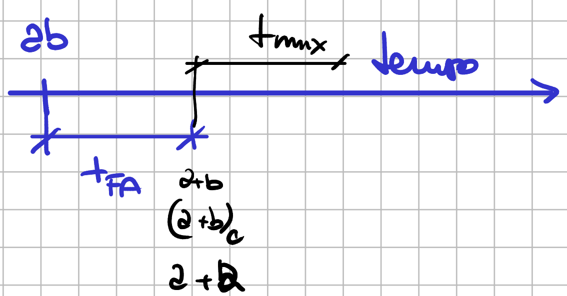
$\lceil \lg_2(2n) \rceil + \lceil \lg_2(2^n) \rceil$

AND OR

$= \lceil \lg_2(2n) \rceil + n$

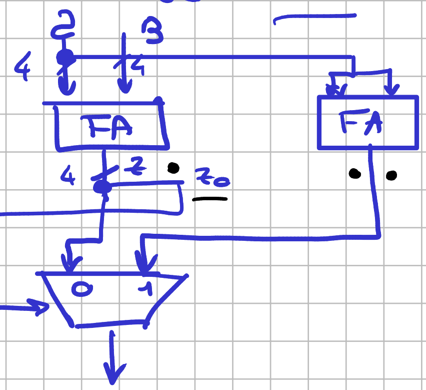


$\max\{a\Delta t, b\Delta t\}$ + t_{mux}



if (a+b è pari) → a+b

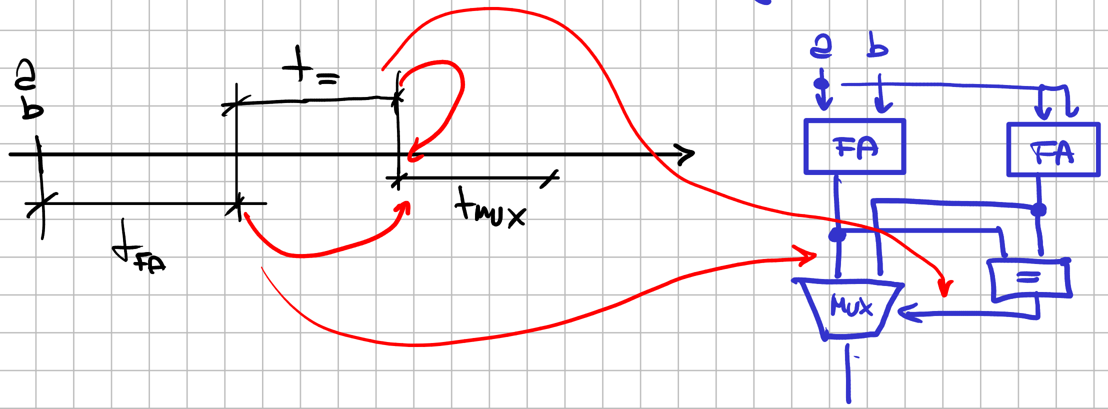
else a+a



t_{FA}

t_{mux}

if(i bit di a+b = a+a) =



$f(x) = \underline{se\ (x\ pari)}\ \{\ se\ x\ \underline{(divisibile\ x\ 4)}\ then\ \underline{z = 1}$

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\quad else\ \ z = 0\ \}$

x 4bit

$\qquad else\ \ \underline{z=1};$

8 4 2 1

x_3	x_2	x_1	x_0	z
–	–	–	1	1
0	0	0	0	1
0	1	0	0	1
1	0	0	0	1
1	1	0	0	1

$\} \ 0-00|1$

$\} \ 1-00|1$

$4 bit \Rightarrow [0, 15]$

$\boxed{\begin{matrix}0\\4\\8\\12\end{matrix}}$

2 livelli di logica

$z = x_0 + \bar{x}_3\,\underline{\bar{x}_1\bar{x}_0} + x_3\,\underline{\bar{x}_1\bar{x}_0} = \underline{x_0 + \bar{x}_1\bar{x}_0}$

$\Downarrow$

upper bound livelli di logica ? (con porte da 2 ingressi)

$2^n\{\ \overset{n}{\sim}|$ z

z = OR di al più (2^n-1) ingressi ciascuno AND di n bit

$$\underbrace{\lceil \lg_2 (2^n-1) \rceil}_{\substack{n \\ OR}} + \underbrace{\lceil \lg_2 n \rceil}_{AND}$$

n=4

$\boxed{4+2=6}$

x_3	x_2	x_1	x_0	z
–	–	–	1	1
0	0	0	0	1
0	1	0	0	1
1	0	0	0	1
1	1	0	0	1

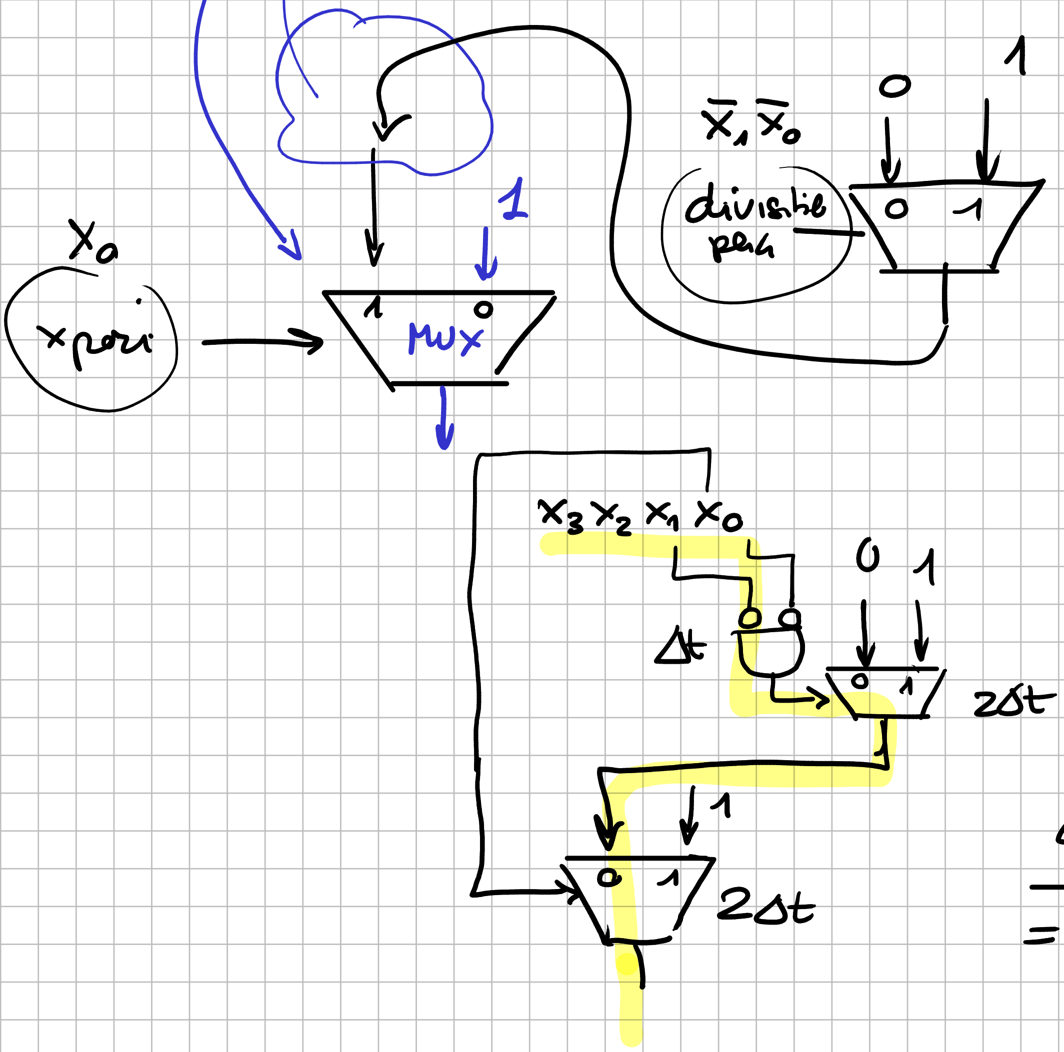
x_3x_2 \ x_1x_0	00	01	11	10
00	1	1	1	0
01	1	1	1	0
11	1	1	1	0
10	1	1	1	0

$z = \bar{x}_1 + x_0$

$f(x) = \text{se } (x \text{ pari}) \{ \text{se } x \text{ (divisibile x 4) then } z = 1 \text{ else } z = 0 \}$

x 4bit

else $z = 1$



$t_{mux} = 2\Delta t$

$$\Delta t + 2\Delta t + 2\Delta t$$
$$= 5\Delta t$$